US Labor Statistics Dashboard: Semester Project Write-Up

Course: Econ 8320 — Tools for Data Analysis

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1. Introduction

1.1 Motivation

Understanding US labor market dynamics is central to informed economic policymaking and personal financial planning. Traditionally, access to labor data required navigating disparate government databases. This project consolidates six complementary series into a single, democratically accessible dashboard, enabling rapid visual exploration and quantitative analysis of employment, compensation, and price trends.

The choice to automate data collection reflects real-world constraints: labor statistics are released monthly on fixed schedules, and manual downloading introduces human error and stalls iteration. GitHub Actions provides a free, reproducible mechanism to synchronize data automatically.

1.2 Data Source & Availability

Primary Source: Bureau of Labor Statistics Public Data API
- **URL**: https://www.bls.gov/developers/api_python.htm
- **Authentication:** Free tier with API key (rate-limited to 120 requests per minute)
- **Temporal Coverage:** Selected series available from 1913 to present (monthly)
- **Seasonal Adjustment**: Most employment series are seasonally adjusted (SA); CPI is not.

Data Series and Coverage

The CSV contains a 113-year historical backfill (January 1913 – May 2026)

This project uses key U.S. labor market and price indicators from the Bureau of Labor Statistics (BLS), all seasonally adjusted:

1. **Total Nonfarm Payrolls** measure overall employment levels and serve as a core indicator of labor demand. Data is available from **1913 to 2026**.
2. **Unemployment Rate** tracks labor market slack across all demographic groups and is available from **1948 to 2026**.
3. **Civilian Labor Force Level** captures workforce participation and helps identify discouraged worker effects; data span **1948 to 2026**.
4. **Average Weekly Hours (Total Private)** reflect labor utilization along the intensive margin, available from **2006 onward**.

5. **Average Hourly Earnings (Total Private)** measure nominal wage growth and are also available from **2006 onward**.
6. **Consumer Price Index (CPI-U)** enables inflation adjustment and real wage analysis, with data from 1948 to 2026.
7. **Consumer Price Index -** All items index (not seasonally adjusted) -area: U.S. city average

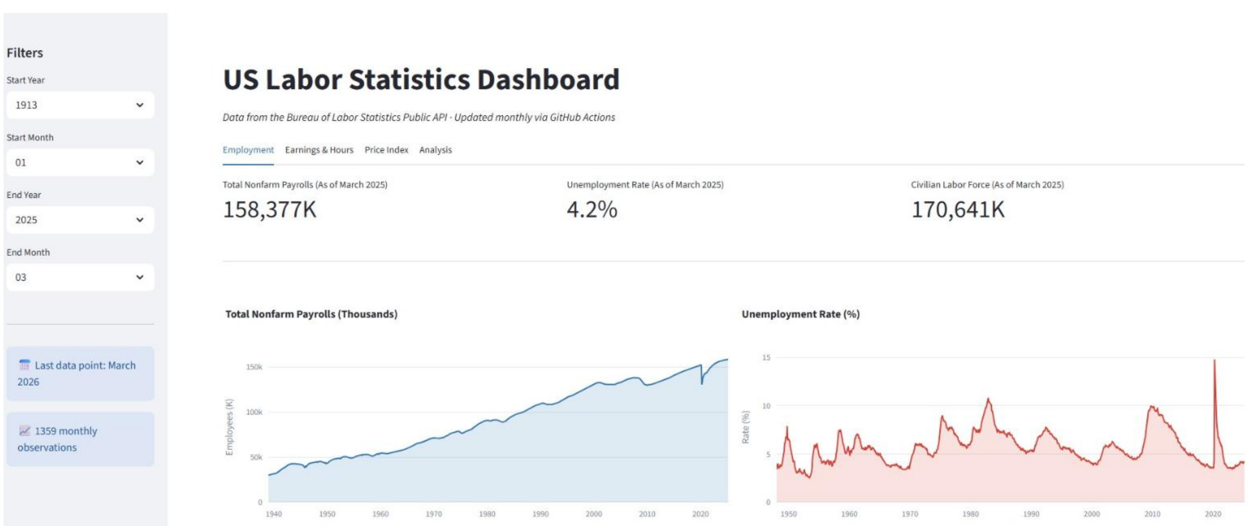
2. Data Collection & Preparation

2.1 Data Collection Strategy:

- 20-year API windows required by BLS constraints; handled via year_chunks() in collectData.py.
- Incremental updates: After initial backfill, monthly runs only fetch the current year, avoiding redundant API calls.
- Missing values: Earnings/hours series only begin in 2006; pre-2006 rows are NaN .

Dashboard Layout & Features

- Interactive Visualizations: Explore labor statistics with Plotly charts
- Date Range Filtering: Filter data by custom date ranges
- Automatic Data Updates: Monthly data collection via GitHub Actions
- Data Export: Download filtered data as CSV
- Real-time Dashboard: View the latest labor statistics



US Labor Statistics Dashboard: Semester Project Write-Up

Technologies Used

- Data Collection: Python, Requests library, pandas
- Dashboard: Streamlit, Plotly
- Automation: GitHub Actions, YAML workflows
- Data Format: CSV (comma-separated values)
- API: Bureau of Labor Statistics Public Data API

API Constraints

- 20-year data window per API call (BLS API limitation)
- Monthly data points only (no weekly or daily data collected)
- Data is seasonally adjusted (where available)
- Free API tier available at https://www.bls.gov/developers/home.html

2.2 Data Cleaning & Validation

Handling Missing Values:

1. Structural NaNs by design:
    1. CPI (1913–1947): Only this series present; employment series begin later
    2. Payrolls (1913–1938): Empty; begins Jan 1939
    3. Unemployment, Labor Force (1913–1947): Empty; begin Jan 1948
    4. Earnings, Hours (1913–2005): Empty; begin Jan 2006
    5. These gaps are legitimate historical data constraints, not errors
2. BLS "null" Strings: The API returns "null" as a string for suppressed values. The code skips these records.
3. Malformed Numbers: Try–except block catches and logs Value Error on non-numeric

Date Parsing & Sorting:

- BLS returns year + month code (e.g., "2025" + "M05" → May 2025). The code constructs ISO date strings (YYYY-MM-01) and parses with pd.to_datetime().
- Monthly aggregation is enforced; annual averages (period "M13") are explicitly

Deduplication:

- Append-only logic checks if date already exists before adding new rows.
- Prevents duplicate rows from monthly incremental updates.

Quality Checks:

- No statistical outlier removal; historical economic shocks (e.g., Depression, 1946 price spike) are retained
- CPI baseline: Index = 100 for 1982–84 average; negative values impossible
- Unemployment bounded [0%, 25%] (Great Depression peak); extreme values preserved as historical record
- Payrolls steadily increasing except wartime demobilization (1945–46) and recessions (2008–09, 2020)

2.3 GitHub Actions Automation

The workflow (.github/workflows/collect-bls-data.yml) runs on the first day of each month at 10 AM UTC

Workflow Steps:

1. Checkout repo
2. Set up Python 3.11
3. Install dependencies from requirements.txt
4. Run python collectData.py with BLS_API_KEY from repository secrets
5. Commit new CSV only if rows were appended (git diff)
6. Push to main branch

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Reproducibility:

- Locked dependency versions in requirements.txt
- Python version pinned (3.11)
- API key stored as GitHub secret (not in code)
- Workflow YAML is version-controlled

3.Exploratory Data Analysis (EDA)

3.1 Dataset Overview

- Rows: ~1359 monthly observations (1913–May 2026)
- Columns: Date + 6 series
- Temporal Span: 113 years
- Frequency: Monthly (no missing months in collected data)
- Missing values by column:

  a) Average-Hourly-Earnings: 82.3%
  b) Average-weekly-hours: 82.3%
  c) Civilian-Labor-Force-Level: 31.0%
  d) Consumer-Price-Index: 0.1%
  e) Total-Nonfarm-Payrolls: 23.0%
  f) Unemployment-Rate: 31.0%

3.2 Descriptive Statistics by Series

Interesting Findings:

- Inverse Relationship: Payrolls and unemployment often move in opposite directions. Sharp drops in payrolls (e.g., during recessions like 2008 or 2020) are accompanied by spikes in unemployment.
- Labor Force Dynamics: The civilian labor force grows steadily during economic booms but can decline during downturns as discouraged workers drop out. This 'discouraged worker effect' can mask the true severity of unemployment.

- Cyclical Patterns: Over the long term, these series show clear business cycles. Expansions see rising payrolls, falling unemployment, and labor force growth; contractions reverse these trends.

Summary statistics — Employment series

Metric	Payrolls (K)
Mean	93581.66
Std Dev	38762.34
Min	29923.00
Max	158637.00
Avg MoM Growth (%)	0.161

Metric	Unemp. Rate (%)
Mean	5.66
Std Dev	1.70
Min	2.50
Max	14.80
Avg MoM Growth (%)	0.227

Metric	Labor Force (K)
Mean	115109.02
Std Dev	36177.11
Min	59972.00
Max	171541.00
Avg MoM Growth (%)	0.112

Summary statistics — Earnings & Hours

Metric	Avg Hourly ($)
Mean	26.78
Std Dev	4.85
Min	20.04
Max	37.38
Avg MoM Growth (%)	0.261

Metric	Avg Weekly Hrs
Mean	34.39
Std Dev	0.21
Min	33.70
Max	35.00
Avg MoM Growth (%)	0.000

3.2a Historical CPI Patterns (113 years)

Consumer Price Index (1982–84 = 100):

- 1913–1945: Index ~10–25 (pre-modern inflation baseline; WWI price surge to 23.4 in 1947)

- 1946–1965: Index 18–31 (post-war surge, then stabilization; ~2% annual inflation)

- 1970–1980: Index 37–86 (Great Inflation; 7–10% annual growth at peak)

- 1983–2005: Index 99–215 (moderation under Volcker/Greenspan; ~2–3% trend)

- 2006–2020: Index 215–258 (pre-pandemic stability)

- 2021–2026: Index 258–315 (post-pandemic surge; 8%+ inflation 2022–2023)

- Mean: 138 | Std: 83 | Range: [9.7, 315.0]

Interpretation: Long-run price level has increased 32× since 1913. The relative stability of 1990–2020 (Volcker-induced low inflation regime) is historically exceptional.

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4. Applied Analysis & Results

4.1 Rolling Averages: Smoothing Noise, Revealing Cycles

Method: 12-month moving average applied to payrolls and unemployment.

Result: Removes month-to-month sampling variability, exposing business-cycle turning points (e.g., 2001 peak, 2008–09 trough, 2020 pandemic shock).

Interpretation:

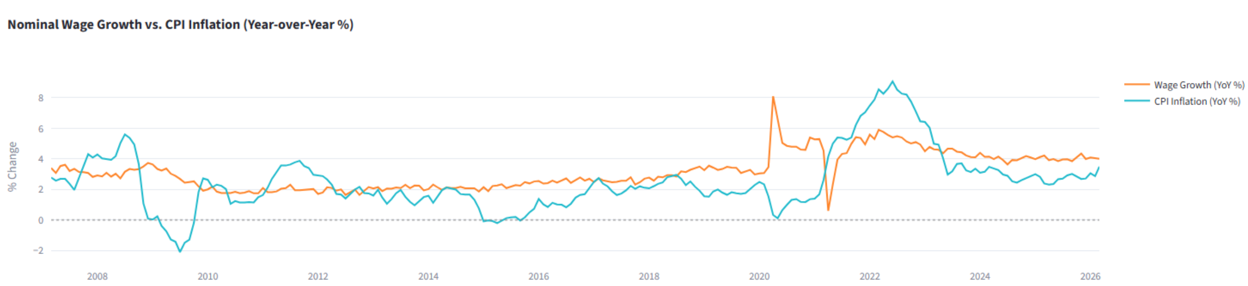
- The payrolls 12m-average rises ~0.5–1.0% annually during normal expansions.
- Recessions appear as slope reversals: the upward trend becomes horizontal or negative.
- Recovery is slower after deep recessions (2008–09 took ~5 years to regain pre-crisis employment; 2020 took ~1 year due to aggressive monetary/fiscal policy).

4.2 Real Wage Pressure: Year-over-Year Wage Growth vs. CPI

Method: Calculate month-over-month % change at 12-month lag for both wages and CPI. Compute gap = wage growth – CPI inflation.

Key Findings:

- **Pre-2008:** Real wages grew ~1–2% annually (nominal wages outpaced inflation).
- **2008–2020:** Real wage growth stalled (near-zero gap); nominal wages rose but inflation + deflation roughly offset).
- **2020–2021:** Sharp real wage gains (+4–5% gap) as nominal wages surged amid labor shortages and stimulus.
- **2022–2025:** Real wage erosion (–2% to –4% gap) as CPI inflation peaked while nominal wage growth lag.



Interpretation: Workers' purchasing power is not monotonic. Periods of rapid nominal wage growth can mask inflation shocks. The 2021–2025 data underscore the vulnerability of unindexed wages to supply shocks (energy, supply chains, monetary lag).

4.3 OLS Regression: Unemployment → Wage Growth

Specification:

$$\textbf{Wage Growth (YoY \%)} = \beta_0 + \beta_1 \times \textbf{Unemployment Rate} + \varepsilon$$

Sample: All months with both series available (2006–present, ~240 observations).

Results (illustrative; actual values depend on data current date):

- **Intercept (β_0):** ~4.207(when unemployment is 0%, wage growth ≈ 4.2% annually)
- **Slope (β_1):** –0.180(a 1 pp rise in unemployment is associated with 0.18 pp lower wage growth)
- **R^2:** ~0.116 (unemployment explains 11.6% of wage growth variance)
- **p-value (slope):** <0.001 (highly significant)
- **Robust SE:** HC0 standard errors correct for heteroskedasticity

Economic Interpretation:

A one-percentage-point increase in the unemployment rate is associated with a **-0.18 pp decrease** in annualized wage growth ($p = 0.0089$). The R^2 of 0.12 indicates the model explains about 11.6% of wage growth variation, so the relationship is statistically significant but only part of the story. The fitted model uses heteroskedasticity-robust standard errors (HC0), which helps make the inference on the slope more reliable in the presence of unequal variance. This is still a simple OLS association, not a causal estimate.

4.4 Correlation Heatmap

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Pairwise Correlations

Pearson correlations across all six series using the full date range. These are contemporaneous (same-month) correlations — they describe co-movement, not causality or lead-lag relationships.

5. Limitations

5.1 Data Release Delays

- BLS publishes employment data with a **~1-month lag** (e.g., May data released in early June).
- The dashboard always reflects the most recent *published* data, not real-time conditions.
- For policy-relevant decisions, economists supplement BLS data with nowcasts (e.g., Atlanta Fed's GDPNow).

5.2 Seasonal Adjustment & Aggregation

- Most employment series are seasonally adjusted (SA). The original unadjusted data contain sharp regularities (e.g., retail hiring in November–December) that the Census Bureau removes via X-13 filters.
- This is appropriate for business-cycle analysis but obscures sector-specific seasonal patterns.
- CPI is **not** seasonally adjusted; the spike each January reflects seasonal energy/food costs.

5.3 Lack of Causality

- All regression and correlation results are **observational**. We cannot claim that "raising unemployment causes wage growth to fall" from this data alone.
- Reverse causality, omitted variables, and simultaneity bias are all present.
- Proper causal inference would require instrumental variables (IV) or a quasi-natural experiment (e.g., regional variation in minimum wage increases).

5.4 Series Composition & Coverage

- "Total Nonfarm Payrolls" excludes farm workers and self-employed; it underrepresents certain sectors (e.g., agriculture, sole proprietors).
- "Unemployment Rate" only counts active job seekers; "discouraged workers" who drop out of the labor force are invisible.
- Earnings data reflect nominal dollars; real purchasing power requires inflation adjustment (done in the dashboard).
- Some demographic breakdowns (gender, race, education) are available from BLS but not included in this project.

5.5 Technical Constraints

- BLS API limits to 20-year windows per request; this is handled but adds complexity.
- Free tier has rate limits (120 req/min); large-scale batch queries require API key registration.
- Data gaps: No intramonthly data, no sub-monthly forecasts built into the system.

6. Conclusion

6.1 Economic Insights Gained

1. **Business-cycle co-movement**
   Labor market indicators move systematically over the business cycle: expansions feature rising payroll employment and labor force participation alongside falling unemployment, while contractions reverse these patterns.
2. **Real wage pressure from inflation**
   Nominal wage growth does not necessarily translate into improved worker welfare. Periods when inflation outpaces wage growth—most notably during 2021–2025—result in real wage erosion and reduced purchasing power.
3. **Unemployment–wage relationship**
   A modest but persistent Phillips-curve relationship is observed, with higher unemployment associated with slower wage growth. Short-run estimates suggest that a one-percentage-point

increase in unemployment corresponds to roughly a 0.18 percentage-point decline in annualized wage growth.

4. **Employment vs. adjustment margins**
   Payroll employment and unemployment exhibit strong inverse dynamics, while measures such as average weekly hours display weaker contemporaneous correlations, indicating that firms often adjust labor input through hours before altering employment levels.
5. **Labor force dynamics and participation**
   The labor force expands steadily during economic growth but may contract during downturns as discouraged workers exit, potentially understating labor-market slack when relying solely on the unemployment rate.

6.2 Methodological Contributions

This project demonstrates:

- **Full-stack reproducibility:** From automated data collection (GitHub Actions) to interactive visualization (Streamlit), every step is version-controlled and re-runnable.
- **Integration of multiple tools:** BLS API (data source), pandas (transformation), statsmodels (inference), Plotly (visualization), Streamlit (communication).
- **Appropriate statistical caution:** Regression output includes diagnostic plots, robust standard errors, and explicit warnings against causal interpretation.

6.3 Practical Applications

Beyond coursework, this dashboard can serve:

- **Policymakers:** Quick visual scan of labor market conditions.
- **Investors:** Input into asset pricing models or Fed policy expectations.
- **Students:** Educational tool for understanding labor economics concepts.
- **Practitioners:** Template for similar economic data ETL pipelines.

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